

MTH205 Analysis in one variable

Tutorial 6

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(1) Prove the following statements.

- (a) If $f : [0, \infty) \rightarrow \mathbb{R}$ is continuous and f is uniformly continuous on $[a, \infty)$ for some fixed $a > 0$ then f is uniformly continuous on $[0, \infty)$.
- (b) If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and there is $p > 0$ such that $f(x + p) = f(x)$, for every $x \in \mathbb{R}$ then f is bounded and uniformly continuous.

(2) Let $p(x) = x^2 - x + 1$ then compute the Bernstein polynomials $B_n(p)$.

(3) For $n \in \mathbb{N}$ define $f_n : (0, \infty) \rightarrow \mathbb{R}$ by

$$f(x) = \log\left(x + \frac{1}{n}\right), \text{ for every } n \in \mathbb{N}.$$

Investigate the pointwise and uniform convergence of the sequence $\{f_n\}$ on $(0, \infty)$. Also, compute its pointwise/uniform limit if it exists.

(4) Show that the sequence $\{g_n\}$ of the functions from $\mathbb{R}$ to $\mathbb{R}$ given by $g_n(x) = \frac{1}{\sqrt{(1+x^2)^n}}$ for $x \in \mathbb{R}$ does not converge uniformly on $\mathbb{R}$.

(5) Let $S \subseteq \mathbb{R}$ and $\{f_n\}$ be a sequence of continuous functions from S to $\mathbb{R}$.

- (a) If $\{f_n\}$ converges uniformly to some function $f : S \rightarrow \mathbb{R}$, then $f(x) = \lim_{n \rightarrow \infty} f_n(x_n)$ whenever $\{x_n\} \rightarrow x$ in S .
- (b) If $\{f_n\}$ is uniformly Cauchy on a dense subset D of S then $\{f_n\}$ is uniformly Cauchy on S .